

Explainable Artificial Intelligence for Time Series: A Comparative Survey

Xinyue Gu, Linxiao Yang, Liang Sun

DAMO Academy, Alibaba Group, Hangzhou, China; Hupan Lab, Hangzhou, China

{guxinyue.gxy, linxiao.ylx, liang.sun}@alibaba-inc.com,

Abstract

As the applications of time series increase significantly, there are growing interests in explainable artificial intelligence (XAI) for time series. Compared with text, image and tabular data, time series is temporal dependent and less intuitive. The unique nature of time series data and special explanation requirements make existing XAI methods fail to be applied directly. In this survey, we review the recent progress in the field of XAI for time series systematically. We first summarize the key challenges of XAI for time series. We propose a new taxonomy to organize these methods into four categories, including perturbation-based, distribution-aware, decomposition-based and attention score. For each class of methods, we review typical algorithms with the discussion of their strengths and weaknesses. We review the evaluation methods for XAI for time series with an emphasis on faithfulness. To further demonstrate properties of different explanation methods, we perform empirical studies in a time series anomaly detection task. We conclude with a discussion of future research directions.

1 Introduction

Machine learning models for time series data are becoming more powerful and complicated. Early time series models such as ARIMA, Exponential Smoothing, and STL [6], are simple and highly interpretable. However, as modern deep learning algorithms [21, 55, 31, 57] provide superior capabilities in capturing long-range dependencies and intricate temporal patterns, they generally lack interpretability.

There is a growing need for insights into why and how time series models, especially black-box models, make their predictions. Interpretability is crucial in high-stake industries, especially regulated domains such as finance, healthcare and power industry. With interpretable tools, users can understand the mechanisms of the models and verify that the model’s decision-making aligns with established rules or common sense. For example, stakeholders need to understand the features influencing stock prices, utility providers want to see how the model interprets temperature patterns

for energy resource planning, and banks deploying fraud and anti-money laundering solutions need to verify the automated decisions are reasonable, clarifying why certain transaction patterns are flagged as potentially fraudulent. In general, these insights are provided through key segments of time series data.

The eXplainable Artificial Intelligence (XAI) refers to methods and techniques designed to make AI systems more transparent and understandable. XAI has developed numerous useful tools such as SHAP [27], LIME [35], Integrated Gradients (IG) [43], GradCAM [37], etc., allowing even novice practitioners to understand and verify predictions, uncover hidden biases, explore mechanisms and beyond. XAI’s applications span across domains including tabular data, images, and natural language. However, these general-purpose XAI tools are not fully compatible with time series data.

The XAI analysis for time series presents unique challenges due to its temporal dimension. Unlike images that can be readily segregated into foreground and background, time series data contains complex patterns (e.g., trends, periodic fluctuations) and dependencies that make interpretation more difficult. From a presentation perspective, XAI methods must reveal both feature importance and how each feature’s significance varies across different time segments. From an algorithmic perspective, operations such as deleting, replacing, sampling, or marginalizing features must carefully consider their impact on the temporal dimension. Time series contexts exhibit complex patterns that are less intuitive than those found in text or image data, making it difficult for users to quickly discern meaningful components. As most XAI methods were originally designed for explaining image and tabular data, applying them directly to time series is often inappropriate. Furthermore, many XAI algorithms [35, 27] impose assumptions such as feature independence, which is generally violated in time series data where temporal dependencies are inherent.

Formally, we summarize key challenges of XAI for time series as follows:

- **Temporal dependency.** Time series data inherently depends on the order of data points. A reversed time series, for example, can change an increase to a decrease, altering the information conveyed. Understanding the temporal dynamics of the context also helps us better capture abrupt shifts and anomalies. Thus being aware

of the temporal distribution is crucial for time series explanation. Yet it is non-trivial, as multi-level periodical cycles, trends and random fluctuations are generally emerging.

- **Joint interaction of time and features.** In time series data, features interact across jointly to determine the outcome. For instance, cumulative effects describe how past events progressively build their influence, and sustained effects take place when specific patterns maintain their significance over consecutive time intervals. In multivariate time series analysis, the explanation become more complex, involving interaction across different features at time steps. Moreover, the interaction can be characterized at different temporal granularities: individual level interaction reveals the microscopic connections and sequences level interaction allows researchers to understand macroscopic patterns.
- **Explanation baseline.** For those models that cannot alter input dimensionality, replacing features is an alternative for removing features. The explicand’s feature values are switched to auxiliary values, defined as baselines, to mimic the effect of “turning features off” [42]. For image attribution, a black image baseline (zero-valued input) is a fair choice [43]. For tabular data, filling with random numbers, means values or other samples’ values are a common practice [42]. Different baseline approaches can yield different explanation results. For time series data, choosing baselines requires special caution. Each data point is not an isolated entity but part of a rich, dynamic system with temporal dependency, temporal interaction and possible time lags. Traditional baseline generation methods frequently fail to capture these nuanced dynamics, potentially introducing out-of-distribution (OOD) data that misrepresents the actual effects of feature removal. We will delve deeper into the OOD issues in Section 2.2.
- **Evaluation of explanation.** Evaluation validates whether the explanation results are truly reasonable. In static data, individual features can be isolated and assessed. But in time series data, features are tightly woven together and changing along with their context. Ideally, a good evaluation method must go beyond merely testing one feature in isolation. It should consider how features are related and how they contribute to the overall model prediction, especially when dealing with complex temporal contexts and noises.

Recently, some survey papers have been proposed for XAI for time series in the literature [36, 8]. They organize the progress of XAI for time series from multiple perspectives, including ante-hoc (inherently interpretable) vs. post-hoc approaches (explanations for trained models), global (overall model logic) vs. local (specific instance behavior) explanations, and model-agnostic (applicable to various architectures) vs. model-specific techniques (tailored to certain models), and discuss XAI approaches for specific time series models and different stakeholder purposes. Yet, significant challenges typically for time series have not been fully discussed. Motivated by these unique characteristics and chal-

lenges, our work aims to review how existing XAI methods specifically address and optimize explanations for time series data. We focus on examining the innovative strategies developed to overcome time series-specific interpretability obstacles.

Formally, we summarize our contributions as follows:

1. We propose a taxonomy of XAI methods for time series (Figure 2), including model-agnostic and model-specific ones. Specifically, we give a detailed introduction to perturbation-based, distribution-aware, decomposition-based and attention scores methods.
2. We systematically discuss how XAI methods for time series handle the typical challenges arising in time series explanation.
3. We provide practical guidelines for applying different explanation methods with an empirical study comparing different methods.
4. We review different evaluation metrics of these explanation algorithms with the discussion of strengths and limitations.

2 Preliminaries

An *explanation* algorithm attempts to highlight why a model makes a prediction, usually in two paradigms, i.e., *feature attribution* and *feature selection*. Specifically, feature attribution methods assign each feature a score, generally in the additive form. Feature selection methods yield a subset of the most important features. In the following, we introduce some widely-used explanation methods that provide explanations for the predictions of any classifier or regressor. Throughout this survey, f denotes as the model to be explained and $x \in \mathbb{R}^d$ as the features of the instance.

2.1 XAI methods for general purpose

Feature occlusion An early attempt in XAI [54] ooccludes different portions of the input image with grey squares as non-informative baselines, and check the drop in probability of the correct class to calculate the marginal contribution. Later, [44] applies a similar occlusion technique to an LSTM model by masking features one by one with a random noise. To avoid OOD, [46] introduces Augmented Feature Occlusion (AFO), that perturbs the feature x_i using bootstrapped empirical distributions. While conducting feature occlusion in an image is straightforward due to the independence assumption, it requires more caution when generating in-distributional baselines for time series data.

LIME The Local Interpretable Model-Agnostic Explanation (LIME) [35] trains a local surrogate model g for explanation. Specifically, LIME generates perturbed samples around the original instance, assigns weights based on proximity to the original point, and then trains a locally interpretable linear model g to approximate the complex model f in that neighborhood. The neighborhood range should be carefully balanced: small enough to allow the surrogate model to linearly approximate the original model with good accuracy, yet large enough to provide sufficient data for training the surrogate

model. Otherwise, the explanation can be unstable or even flipping the direction.

The challenge of applying LIME to time series lies in defining an appropriate neighborhood. Treating each time step’s feature independently for perturbation may not be reasonable to create a meaningful local linear approximation.

SHAP SHAP (SHapley Additive ExPlanations) [27] explains model predictions using Shapley Value (SV) from the game theory perspective. It measures each feature’s average marginal contribution to the predictive outcome over all possible feature permutations. Theoretically, computing the exact SV requires solving 2^d conditional equations (where d is the number of features), making it computationally intractable for high-dimensional data. To reduce the computational cost, KernelSHAP [27] fits Weighted Least Squares (WLS) with a subset of the feature combinations.

Baseline selection and computational complexity are the two key challenges for applying SHAP to time series.

As most of the models only accept input of original length, practically SHAP uses feature replacement instead of feature removal. The values used for replacement are baselines. SHAP usually replace the features with zeros, means or random values. Variants include BSHAP [42] that imputes values of other reference samples as baselines and [13] that samples from conditional distribution. However, properly replacing features may introduce OOD inputs to the model, while conditional distribution may generate bias in estimation.

Besides, the computational complexity issue is more pronounced in time series data, as the time dimension multiplies the number of features to be permuted. Whether the vanilla KernelSHAP still works in this context requires further investigation.

Masking Masking originates from computer vision [12]. It aims to find the subregion of input image that leads to the significant change in model performance after feature replacement. Formally, it formulates an optimization problem to determine the masked positions m .

Let $\Phi(x, m)$ represent the value of input x after applying mask m , where m is a mapping from x to $\{0, 1\}$, and the entry u in x denoted as x_u . If $m(u) = 1$, then $\Phi(x, m)_u = x_u$; otherwise $\Phi(x, m)_u$ is replaced according to different policies (constant/noise/blur). The optimization aims to learn the smallest mask m (i.e., $\|1 - m\|_1$) such that $f(\Phi(x, m))$ deviates from $f(x)$ as much as possible, while minimizing the total-variation norm of m (i.e., $\|\nabla m\|_\beta$). This approach seeking the smallest significant mask is called a “deletion game” [12]. Alternatively, a “preservation game” retains the smallest subset of features that preserves the original prediction with minimal change. **Gradient-based** Gradient-based attribution methods explain predictions by analyzing how small input changes affect outputs. Vanilla gradients [39] use partial derivatives as attribution scores, while multiplying gradients with inputs, as in first-order Taylor approximations [38], improves results. GradCAM [37] applies the multiplication idea to hidden layers. DeepLIFT [38] computes feature importance by comparing input activations to reference activations and propagating differences. Integrated Gradients (IG) [43] compute the importance of each feature by integrating the model’s gradients along a path from a base-

line input, satisfying axioms of attribution and showing applicability to various domains.

2.2 Potential Issues in Time Series

Applying general-purpose XAI methods to time series data introduces additional complexities. Figure 1 illustrates two key challenges: 1) OOD issues (subfigure a); and 2) difficulties in capturing interactions (b and c). Note that Figure 1.a represents the original feature curve.

Out-of-distribution issues Feature replacement or feature sampling in XAI renders discussions of OOD issues unavoidable. The ideal replacement should serve as a neutral baseline without introducing new information. However, no replacement method perfectly mimics feature removal. Zero or fixed values may introduce outliers. Mean replacement is generally safe for continuous variables, but can be meaningless for categorical variables. In-sample value replacement (such as BSHAP) can create improbable samples, e.g., females with long beards.

For time series explanation, feature replacement should meanwhile preserve temporal dependencies. We demonstrate the OOD challenges in time series in Figure 1.a with a univariate time series example. Random value replacement (a.2) violates temporal smoothness and create serious OOD issues. Mean value replacement (a.3) overstates marginal contributions by suggesting artificially low baselines. Phase-shifted replacement (a.4) misrepresents the temporal patterns of the original data. To address OOD issues for time series, we review temporal distribution-aware approaches in Section 3.2.

Complex interaction patterns For time series interaction, one challenge is identifying the relationships along the time dimension and between features across time periods.

For instance, consider two features (blue and green lines in Figure 1.b) that jointly influence the output but peak at different times. The subplot b.1 represents two features in a dataset where their joint interaction may originate from different time windows. For example, the second peak of the blue line and the second peak of the green line occur at different time points, but they jointly contribute to the outcome of interest. Traditional methods that analyze interactions within the same time window (Figure 1.b.2, yellow shade) miss these critical cross-temporal relationships. Capturing such interactions requires more flexible methods, which attention score in Section 3.4 may excel at this task.

The interactions in Figure 1.c span multiple consecutive time steps. The subplot c.1 contains a temporal interaction of a series of consecutive time steps. A common mistake is in c.2 that only captures a subset of the time steps, missing the broader time period. We further explore the interaction issues in the context of “deletion” and “preservation” via toy examples in Figure 3 in Section 3.1. Due to the correlation along the timeline, the perturbed values for time series can also be generated through inferring from the existing data, which we will introduce in Section 3.2.

2.3 Discussions

We conclude this section by discussing the improvement direction for these aforementioned explaining methods when

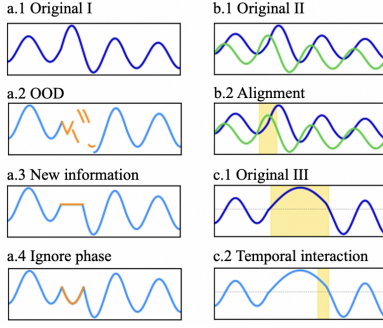


Figure 1: Illustration of key challenges in time series XAI. a: OOD issues in feature perturbation. Original time series compared with three problematic replacement scenarios – (a.2) randomly sampling violates temporal smoothness, (a.3) mean filling introduces misleading baselines, and (a.4) phase-shifted values misrepresents temporal patterns. b: Feature interaction across time: Original two features (b.1) show joint interaction, which is missed when analysis is restricted to simultaneous time windows (b.2). c: Consecutive time points as interaction: Complete anomaly pattern is yellow-shaded, above grey line (c.1), compared with incomplete interaction identification (c.2), where the deletion game finds only a subset of the period still making the objective drop.

applied to time series. We then present our taxonomy based on possible improvement directions.

Improvements for time series. The mainstream methods introduced in Section 2.1 involve perturbation or sampling without requiring knowledge of model structures. For such model-agnostic XAI methods, their adaptation to time series focuses on two key considerations: where to perturb and what values to replace. In particular, when determining perturbation locations, computational efficiency is a major concern, as perturbing all the features at all time steps would be prohibitively expensive. The “deletion” and “preservation” methods discussed in 2.1 have great potential to be extended to time series thanks to their flexibility in selecting perturbation subsets. Meanwhile, the choice of replacement values directly impacts the ability to distinguish between temporal context effects and the actual signal.

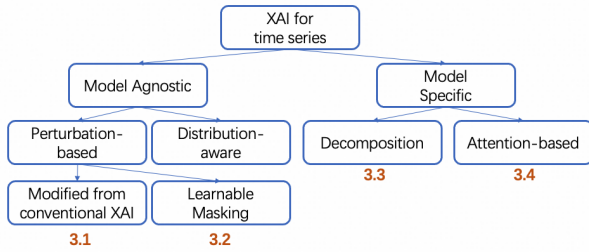


Figure 2: Taxonomy of XAI methods for time series.

Taxonomy. Our taxonomy of XAI methods for time series is shown in Figure 2. As summarized in the last paragraph, XAI methods for time series can be categorized into those derived from or motivated by classic post-hoc XAI methods for general purpose; and those intrinsically designed for time series models. We term them as “model-agnostic” and “model-specific” methods, respectively.

Within model-agnostic methods, gradient-based methods do not suffer from exponential increases in computational

cost, thus they are often applied directly without modification when applied to time series. In contrast, feature replacement approaches should be tailored for time series adjustments.

We categorize model-agnostic methods into two groups: perturbation-based approaches (Section 3.1), which focus where to apply replacement, and distribution-aware methods (Section 3.2), which focus on what values to use for replacement by extensively modeling time series distributions. Section 3.1 will examine SHAP-based and LIME-based modifications for time series, as well as learnable masking techniques that leverage temporal properties.

Model-specific methods do not follow a unified paradigm. Some incorporate interpretable parameter weights, while others leverage specific network modules as interpretable components. Given their variety, we will focus on the most prominent approaches: decomposition in Section 3.3 and attention-based methods in Section 3.4.

In Section 5, we will evaluate the effectiveness of mainstream methods in time series and also include gradient-based methods for comparison.

3 XAI for Time Series

In this section we discuss XAI methods for time series in detail. Section 3.1 and Section 3.2 introduce model-agnostic methods and Section 3.3 and 3.4 introduce model-specific methods.

3.1 Perturbation-based Methods

The most adopted explanation methods are through perturbation, testing how an output responds to a change in input.

SHAP-based Methods Due to its good adaptation and user-friendly implementations, SHAP becomes the basis for a few explanation methods for TS. A straightforward method is to treat each feature at each time step as an independent variable and perturb them individually. Evidently, this naive approach suffers from several drawbacks. As studies [48, 47] have pointed out, while SHAP and its variants (such as KernelSHAP, DeepSHAP, TreeSHAP, etc.) assume feature independence, this assumption often violates for time series. Besides, the long input windows involved in TS predictions add burden to the computation complexity of SHAP. To address the above two challenges, studies on adapting SHAP to TS primarily focus on how to **a)** incorporate the temporal dependency; **b)** group the permutations; and **c)** decrease the computational cost.

First, temporal dependency can be incorporated through adjusting the regression models in KernelSHAP. [47] follows this idea and proposes VARSHAP that replaces KernelSHAP’s linear model by time series models, including autoregressive models, moving averages, etc.. Yet VARSHAP still considers the same feature at different time steps as different players, from the perspective of game theory. Inspired by the property of *time consistency* [34], [47] proposes an optimized version of KernelSHAP dubbed Time Consistent SHAP, which first computes SHAP of each feature and then allocates SHAP for time steps as subgames.

Second, as for grouping the permutations, [16] proposes to perturb the current state and freeze the rest. The limitation of

this approach is that it can only make explanation within current state. An alternative is to define sequence-wide perturbations. TimeSHAP [2] extends KernelSHAP and groups sequential data into coalitions, so that the granularity of explanation extends to feature-level, time-step-level and cell-level. Perturbing coarse level inputs also helps reduce the computational complexity though, the assumption in TimeSHAP that recent sequences exert greater influence on the outcome may violate the truths.

Third, for the computational efficiency, [2] proposes a pruning strategy that searches for a starting point of the calculation and then disregards the past series with small SHAP values. [47] proposes the Time Consistent SHAP to reduce the computational cost from exponential growth to linearly scaling with the size of time steps.

LIME-based Methods The process of LIME generally involves generating perturbed values in the neighborhood and training a local model. In adapting LIME to the TS data, there are three key problems which require careful consideration [41], including: a) what to explain, namely a meaningful interpretable representation of a TS; b) how to perturb a TS which respects the temporal coherence of the data; and c) where to perturb, i.e., the definition of a locality for perturbation and model training.

To address the three pivotal questions above, [41] provides a typical solution. It proposes a meaningful conceptualization via a segmentation algorithm to identify change points of TS. To perturb a TS, the study utilizes harmonic analysis to find realistic background content as the perturbation patterns. Finally, to define the local neighborhood, dynamic time warping (DTW) is employed to measure distance between two TS.

Learnable masking SHAP-based methods generally assign alternative values to random data points. Unlike tabular data, the tremendous TS data volume makes it impossible for all data to be perturbed, so studies expect only a group of selected features to be masked. For TS problems, the core challenge is two-fold. The first challenge is the selection of the masked part. Second, how to build uninformative perturbations while keeping data in distribution. For TS data, in order to preserve the temporal dependency, the replaced value should not only be in the trends and cycles [7], but also avoid label leakage [19].

Dynamask [7] is an early study introducing the “preservation” mode that minimizes the masked loss by inspecting the effect of a dynamic perturbation operator. In order to find perturbations that are in-distribution, it introduces three blurring techniques to generate context-adapted in-distribution perturbations, which are a Gaussian blur [12] and two convex combinations of the past and the future. Dynamask is the pioneer that raises a wise choice for TS perturbation, but values using temporal Gaussian blur are fixed and local. [11] claims that it fails to reveal long-term dependencies of multivariate TS, particularly repetitive patterns. To overcome this drawback, [11] generalizes Dynamask and replaces perturbations with learnable neural networks. This generalized approach also presents the results for both “deletion” and “preservation”, showing that deletion underperforms on most metrics. Figure 3 provides two toy examples of deletion and preser-

vation, highlighting their strengths and weaknesses. Overall, preservation generally has a more straightforward objective function, but it may miss certain conditions when any of the conditions leads to a particular outcome. On the other hand, deletion only needs to disrupt the original pattern. Thus, it may overlook important variables where multiple key data points interactively contribute to the outcome jointly.

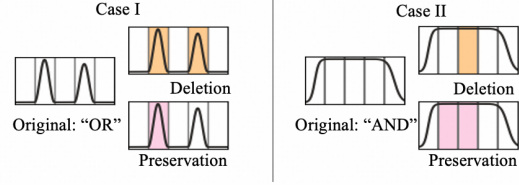


Figure 3: Strengths and limitations of deletion and preservation masking methods in anomaly detection. Case I (“OR” Relationship): both peaks indicate anomalies, but preservation only detects one time step; Case II (“AND” Relationship): slight deletion disrupts the interaction and only finds one anomalous time step, while complete preservation accurately identifies the anomaly.

The above studies of perturbations still risk label leakage, making the altered values “informative”. To resolve this, [26] introduces counterfactual samples to build uninformative perturbations in classification problems in “preservation” framework. This also brings a beneficial byproduct, which extends instance level saliency maps to inter-sample level. Moreover, a sparse stochastic gate [51] replaces the original hard mask mechanism by using noise to approximate the Bernoulli distribution in the masking process. This replacement helps to smooth the TS and lower the variance, while maintaining the data distribution.

3.2 Distribution-aware Methods

Section 3.1 explores the perturbation-based methods that select locations for perturbation. This section examines distribution-aware methods that determine replacement values in the perturbation process. Specifically, distribution-aware methods analyze features’ temporal distribution and measure output discrepancies from in-distribution input alterations. The premise is that feature contributions arise from distributional changes. So these methods aim at removing contextual information.

Feature Importance in Time (FIT) [46] is a typical explanation method that quantifies feature contribution by analyzing temporal shifts. By generating counterfactual inputs within feature distribution, FIT computes the excess premium of temporal distribution shift (current shift) deducted by unexplained distribution shift (learned counterfactual shift). In this way, it quantifies the marginal contribution of the current input which the counterfactual values cannot make. The shifts are measured via KL-divergence between the original and the masked conditional probabilities of the outputs. Building on this approach, [23] proposes Windowed Feature Importance in Time (WinIT), which uses prediction difference and Jensen-Shannon divergence as distance metrics. [28] extends the methodology by learning the possible sets of within-distribution inputs through conditional variational

autoencoder (CVAE). More recently, [53] highlights the critical yet often overlooked inter-observation interactions in time series, introducing FDTempExplainer, a method using in-distribution alternation and functional decomposition to reveal joint interaction of time and features across varying temporal scales.

Critically, distribution-aware methods can strategically mitigate the OOD issue by:

1. Generating counterfactual inputs within feature distributions, then measuring distribution shifts using probabilistic divergence metrics;
2. Accounting for temporal dependencies at earlier time steps and interactions;
3. Explaining feature contributions without introducing statistical implausible feature combinations.

3.3 Decomposition-based Methods

Decomposition is a typical approach to analyze time series, and the underlying structure revealed by decomposition can serve as an explanation. This subsection covers several typical decomposition-based methods that utilize transformation or aggregation of the model units.

[30] proposes Contextual Decomposition (CD), an approach that decomposes standard LSTM into phrase-level contributions. By linearizing activation functions, CD distinguishes between contributions made solely by the current phrase and other phrases, leading to more accurate identification between positive and negative sentiments. The encouraging performance motivates further XAI research along this line of model decomposition, including Agglomerative Contextual Decomposition (ACD) [40], IMV-LSTM [15], REAT [10] and a ReLU-activated Transformer [52]. Specifically, ACD extends CD scores from LSTM to arbitrary DNNs by devising linear decomposition strategies for convolutional, max-pooling, ReLU and dropout layer. ACD employs an agglomerative hierarchical clustering procedure to automatically select and combine tokens, and enables explanation on different granularities. A further generalization to RNN, called REcurrent ATtribution Method (REAT) [10], gives attributions on GRU, LSTM and their bidirectional versions. REAT only reveals patterns learned by RNNs, but also has the debugging ability to detect vulnerable data. [52] is specially designed for ReLU-activated transformer. Their ReLU decomposition separates the additive biases in the linear layers and mitigates the issue of accumulated errors that occurs in ACD. While the approaches above are not specifically designed for time series, they provide insights into how to decompose time series models into more interpretable cognitive chunks. However, these techniques mainly focus on a univariate series and may not be straightforward to apply to multivariate features evolving over time. To address this issue, [15] proposes an Interpretable Multi-Variable LSTM (IMV-LSTM) to calculate variable-wise temporal importance, allowing for a comparative analysis of the importance of the time dimension and the feature dimension. Furthermore, by leveraging the properties of LSTM and the Expectation-Maximization (EM) algorithm, it efficiently computes the time and feature dimensions simultaneously.

3.4 Attention Score

Inherent explainable deep learning model provides interpretability by designing cognitive chunks [9], or the basic units of explanation. Some typical works include [32, 20, 4]. Due to the diversity in this category, in this survey we primarily focus on the most mainstream subset, i.e., the attention mechanisms. More details on the discussion of attention for general XAI can be found in [3].

Attention mechanisms are good at capturing long-range dependencies and have wide applications in TS modeling [49]. Fortunately, the mechanisms also provide explicit interpretation ability for models, as the attention weights can be used to infer the relationships between output and the given cells in the network. For TS, it is a typical practice to design a structure with attention weights to the underlying features and time steps as the inherent importance attribution.

The crux of achieving interpretability in attention mechanisms is to determine what information should be conveyed to the attention. RETAIN [5] uses a two-level neural attention, the first level for individual clinical visits and the second level for specific clinical variables within those visits. The attention is applied in reverse chronological order starting from the most recent visits and moves backward. This aligns with how clinicians often review a patient’s history. The Input-Cell Attention (ICA) [17], on the contrary, addresses the issue of vanishing saliency, avoiding inputs to become less discernible over time or through the layers of the network. By easing the vanishing gradient problem, the estimation of the importance of each input feature is more robust. Temporal fusion transformers (TFT) [24] incorporates even more various types of relationships including global, temporal dependency and events. Each attention head can potentially learn different patterns. Also, TFT encodes static covariates separately from time-varying inputs. Thus, the model can interpret non-time-varying variables (such as a store’s location in retail sales forecast) to the prediction. Other interpretable attention mechanisms design for time series include SSD-Net [25] which learns a sparse pattern and seasonal trend, ProTran [45] which learns driving factors, interactions and probabilistic insights. The implementation of attention for Temporal Convolutional Neural Networks (TCN) [33] and LSTM [14] shows the possibility of achieving promising performance while maintaining model transparency.

Attention-based frameworks are inherently well-suited to capturing joint interactions of features and time. Channel Attention and Feature Orthogonalization (CAFO) further enhances the potential of this framework. Unlike TFT’s variable selection network and RETAIN’s retrospective attention, CAFO explicitly employs channel attention and QR decomposition-based orthogonalization to capture the dynamic relationships between features across temporal dimensions.

Despite of the modeling advantages attention enjoys, the fidelity of attention-based mechanisms still remains debated. [18] first shows that the weights are not able to capture key factors, as performance does not drop when these weights are scrambled. The follow-up work [50] argues that such adversarial models may not exist, as parameters in the previous tests are not directly trained and in practice it is difficult to

learn a distant adversary. Instead, their study with alternative tests shows that attention weights can still offer valuable insights.

4 Evaluation Methods

In this section, we review the evaluation methods for XAI for time series in the literature. Among these evaluation methods, most focus on faithfulness and robustness [56, 41]. Faithfulness means the identified features are truly important. Different evaluation methods are employed depending on the availability of ground truth feature importance. Robustness largely ensures that the explanation is consistent when confronting minor input variations.

4.1 Evaluate faithfulness with Ground Truth

When the ground truth is available, the evaluation can be performed in qualitative or quantitative manners. In qualitative analysis, humans typically examine the identified important features through saliency maps and their attribution scores, rankings and signs, to check their alignment with expectations. Quantitative analysis is similar to prediction evaluation by comparing the explanation with the ground truth. Metrics measuring their discrepancy include precision, recall, and many other evaluation metrics calculating areas under different curves, including area under precision (AUP), area under recall (AUR), area under the precision-recall curve (AUPRC), and area under the receiver operating characteristic (AUROC). Areas are more robust to the original point metrics when the ratio of significant features to insignificant features is imbalanced. Metrics used for ranking are also used. A typical example is mean rank, which is the average rank of the feature importance score, selected by the positive ground truth observations.

Some recently proposed evaluation methods for time series include mask information and mask entropy [7]. Specifically, mask information to a mask M and a subsequence $(x_{t,i})_{(t,i) \in A}$ measures the quantity of useful information, which is associated to saliency scores:

$$I_M(A) = - \sum_{(t,i) \in A} \ln(1 - m_{t,i}), \quad (1)$$

where $m_{t,i}$ represents the importance of feature i at time t for the model f to produce the prediction. Since mask information fails to distinguish between certain special cases¹, mask entropy is defined to handle the issue:

$$S_M(A) = - \sum_{(t,i) \in A} m_{t,i} \ln m_{t,i} + (1 - m_{t,i}) \ln(1 - m_{t,i}). \quad (2)$$

Note that mask information and mask entropy are analogous to Shannon information and Shannon entropy, respectively.

¹The authors illustrate that a subsequence with 3 important features can have the same entropy as another subsequence with evenly distributed importance.

4.2 Evaluate faithfulness without Ground Truth

When the ground truth is unknown, it is a common practice to measure the feature’s importance by checking the performance change led by the corresponding feature. Specifically, the performance change can be measured in three different paradigms: deletion game [11, 29] and preservation game [11]. The deletion game removes the most important observations and often measures the performance with accuracy, the drop of cross entropy, the change in predicted class probability (also called comprehensiveness) and the change in absolute prediction. The preservation game is the so-called “performance deterioration test”, which measures the drop in the predictive performance of the black-box model when removing some of the features. It only keeps the most salient features and mostly computes the change of the predicted class probability compared to the original one. SDF is the minimal number of features that are necessary to be perturbed to change the classifier’s output.

4.3 Evaluate robustness

The key to a robust explanation is that similar inputs should lead to similar explanations [1]. The robustness of time series explanation can be evaluated from different primary perspectives, including robustness against noise and non-stationary contexts.

The robustness against noise is a main focus for XAI in tabular data [20], and it is equally important for time series. To this end, [41] reports the proportion of time series with explanations that remain invariant after the time series are perturbed with random noises.

Robustness against non-stationary contexts is also a significant concern specially in XAI for time series. This evaluation is typically conducted by generating non-stationary datasets for evaluation, such as data generated by Hidden Markov Model [46, 23], financial data [24] or simulated aircraft data with temporal variations at different altitudes [22].

5 Experiments

In this section, we present our empirical studies. We emphasize that our goal is not to provide a comprehensive empirical study of explanation methods in different settings of datasets and tasks. Instead, we would like to highlight properties of different methods, and provide guidelines for how to apply them in practice.

For the purpose of demonstration and clarification, we select time series anomaly detection tasks. In real-world applications, explanations for anomaly detection have gained significant popularity because they are often easier to interpret and verify compared to explanations for forecasting. XAI for time series forecasting tasks is more complex because of the continuous ranges of outcomes and the dynamic nature of time series features. Due to the lack of ground truth for real-world datasets, we generate synthetic datasets with known ground truth to evaluate and compare different explanation methods. We consider three major types of temporal feature-outcome relationships, including the ‘AND’ interaction, the ‘OR’ interaction and non-stationary trends.

Table 1: Representative evaluation metrics on faithfulness of XAI methods for time series.

Metric	Favors	✓ Can/ × Cannot	Adopted in
Truth required			
↑Precision/ AUP	preservation game	✓ misidentified unimportant features	[11] [23] [10] [7] [26]
↑Recall/ AUR	deletion game	✓ omitted important features	[11] [7] [26]
↑AUPRC	-	✓ imbalanced salient features	[46][23]
↑AUROC	-	× less sensitive than AUPRC	[46] [23]
↑mask information	-	× sharpness	[11] [7] [26]
↓mask entropy	-	✓ sharpness	[11] [7] [26]
↓Mean rank	-	✓ single cause	[23]
Truth absent			
↓Accuracy	deletion game	✓ multiple anomalies	[11] [7] [26]
↑Cross entropy	deletion game	× probability drop	[11] [7] [26]
↓Sufficiency	preservation game	✓ interaction	[11] [7] [26]
↑AOPC/ Comprehensiveness	deletion game	✓ multiple anomalies	[41] [11] [10] [7] [26]
↓AUROC	deletion game	× imbalanced salient features	[46]
↑prediction	deletion game	✓ multiple anomalies	[23] [29]

5.1 Settings

Baselines. We compare multiple state-of-the-art model-agnostic and model-specific explanation baselines under anomaly detection tasks. The explanations are generated for the standard LSTM classifier in order to compare the model-specific explanations from Contextual Decomposition (CD) [30]. The model-agnostic methods include: *i)* three general explanation methods, integrated gradient (IG) [43], GradSHAP [27], and DeepLIFT [38]; *ii)* two feature occlusion based methods, feature occlusion (FO) [44] and augmented feature occlusion (AFO) [46]; *iii)* one mask based method dynamicMask [7]; and *iv)* two distribution-aware methods Feature Importance in Time (FIT) [46] and Windowed Feature Importance in Time (WinIT) [23]. For WinIT we set the window size to 10, and for all explanation methods involving generators we set the number of generated samples to 5.

Metrics. The baselines assign each test sample an importance score for every feature in every time step. We assess how well these scores coincide with the ground truth (if known) and how these feature-time step pairs are related to the model’s output. For explanation with ground truth available, we report accuracy, recall and precision for the top- k feature-time step pairs as well as the area under the precision-recall curve (AUPRC). For evaluation without ground truth, we conduct ‘deletion’ and ‘preservation’ methods which respectively replace the top- k feature-time step pairs of the highest and the lowest values, with non-informative values. We report the reduction of performance, in terms of AUPRC and accuracy. If these pairs are indeed important, removing them should significantly deteriorate the model performance.

Data. We conduct experiments on synthetic datasets designed for various anomaly detection tasks. Each dataset consists of time series data with three features and a length of 40, represented as $\{x_{ti}\}_{t \in \{1, \dots, 40\}, i \in \{1, 2, 3\}}$. The features are generated from a standard normal distribution $\mathcal{N}(\mu = \mathbf{0}, \sigma = I)$ unless otherwise specified. The anomaly rates are maintained within the range $[.4, .5]$. The specific types of anomalies are as follows:

1. “AND” interactions. In this setting, multiple conditions

collectively determine an anomaly.

- 1) *Consecutive high values.* This represents a straightforward “AND” relationship. Samples with any consecutive 3 points in $\{x_{t1}\}$ that are $\geq 6\sigma$ are classified as anomalous.
 - 2) *Consecutive high values followed by a low value.* In this case, any occurrence of three consecutive high values followed by a point smaller than -6σ is considered anomalous. This scenario tests whether the algorithm can differentiate between mere high-value occurrences and those that lead to an unpredictable shift from expected behavior, addressing issue c.2 in Figure 1.
 - 3) *Stairs.* A one-dimensional variable x stays (almost) constant until a random time when it changes by ± 1 . A variable y of the same length takes the value 1 at the change point and 0 elsewhere. The anomaly results from both the change point and the previous value. This scenario detects whether the algorithm can avoid bringing new information as discussed in Figure reffig:stair and a.3 in Figure 1.
2. “OR” interactions. The explanation methods should identify all the anomaly contributors.
 - 1) *Presence of a high value.* Samples with any time step in $\{x_{ti}\}$ valued $> 6\sigma$ are anomalous. This represents a naive “OR” relationship, revealing especially the efficacy of deletion and preservation mechanisms discussed in Figure 3.
 - 2) *Presence of an extreme value.* Samples with anywhere in $\{x_{t1}\} \geq 6\sigma$ or $\leq -6\sigma$ are anomalous. This interaction is designed to test appropriate masking replacements.
 3. Non-stationary trends. Normal data not only fluctuates within the original Gaussian distribution but also exhibits an additional trend, where anomalies are 6σ larger than the trend.
 - 1) *Periodical trend.* We incorporate the term $\sin(2\pi t)$, $t \in [-1, 1]$ onto the original Gaussian. Wrong ex-

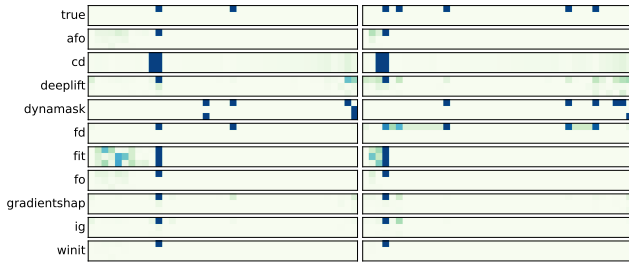


Figure 4: Visualization of saliency maps (highlighted in dark blue) of two representative samples in the OR1 task.

planations may occur if the methods fall into the phase ignorance problem of a.4 in Figure 1.

- 2) *Steadily upward trend.* We introduce an upward line from -1 to 1 onto the original Gaussian. OOD issue may arise from improper masking strategies where the replacement values do not align with the trend, c.f. the case of a.2 in Figure 1.

We randomly divide 3,000 samples into training and test sets with the ratio of 80 : 20, respectively. To compare intrinsic explanation methods with other approaches, we trained LSTM and TFT models on the training set for binary classification.

5.2 Results of Explanation Methods

We report the accuracy and AUROC drop. The accuracy refers to the proportion of true top- k important indices correctly identified among the top- k indices explained by each XAI method. The AUROC drop and difference in prediction reflect the changes observed before and after masking the top- k indices. Besides reporting these metrics, we also present visualization saliency map for in-depth analysis.

As deletion evaluation in Table 2 summarizes, the best-performing methods for 3 AND tasks are FDTempExplainer, IG and WinIT. FDTempExplainer excels in “AND1” by efficiently identifying the interactions along the timeline. In “AND2”, where the anomalous points consist of both high and low values, perturbation-based method do not effectively generate non-anomalous sequences, whereas gradient-based methods have a distinct advantage. In “AND3”, perturbation introduces new steps, and methods like WinIT that considers after-effects are more effective. FO performs the second best, as sampling from uniform distribution occasionally eliminate steps, unlike sampling from $p(x)$ that creates new steps.

For OR tasks, most methods produce an AUROC increase after “deletion”, which is atypical. Similar phenomenon occurs in Trend tasks. Saliency maps in Figure 4 imply differently. GradientSHAP and Dynamask respectively capture the most key features in the left and right figures, but Dynamask appears to be less robust. WinIT always highlights only the first important features. These observations are aligned with preservation results on the right side of Table 2, suggesting deletion’s inadequacy for evaluating OR tasks.

For trend tasks, distribution-aware methods such as FIT and FDTempExplainer effectively capture strong trends. Dynamask struggles to identify distribution information for periodic patterns due to its blurring method.

Our experimental results show that no single explanation method can be universally superior, as their performance varies across different tasks. In the next section, we will provide a brief guideline for selecting the most appropriate explanation method based on the specific characteristics of the data and the task.

6 Practical Guidelines

In this section, we briefly provide some practical guidelines in applying explanation methods for time series. We also comment on the evaluation methods for their strengths and weaknesses, offering insight to select the appropriate methods to meet specific challenges.

Table 1 serves as a short lookup table for evaluation metrics. The first column lists evaluation metrics and their preferred directionality—whether higher or lower values indicate better performance. We categorize metrics suitable for scenarios with and without ground truth. The second column highlights potential statistical biases inherent in these metrics that might preferentially favor certain explanation methods with masking, suggesting potential unfairness when using these metrics for evaluation. The third column details the strengths and limitations of each method, guiding the reader in selecting appropriate ones for specific challenges. The final column provides references in the literature that has employed these evaluation methods.

Choice of metrics. When ground truth is unknown, methods removing important features give high priority to features disrupting the original patterns, and are prone to overlook important features that are not removed. In contrast, methods that only keep important features are more intuitive as they justify the importance of retained features directly. Generally, the evaluation by retaining is better suited for cases with interaction exploration, while removing-based evaluation is more effective at detecting whether all multiple causes are identified, particularly in anomaly detection. It’s important to note that since performance deterioration tests naturally resemble the deletion and preservation game in masking explanation methods, these evaluations tend to favor explanation methods that align with their verification process.

Choice of replacing values. Feature removal can introduce several challenges. For instance, [10] observes in natural language processing (NLP) tasks that removing certain irrelevant words or phrases can lead to significant probability drops, possibly due to the creation of grammatical or syntactic errors rather than actual semantic changes. This makes it difficult to determine whether a removed component is truly essential.

Moreover, not all models can accommodate feature removal seamlessly; most models require inputs of fixed lengths, necessitating replacement values for removed series segments. Common approaches include using means or zeros as replacements. However, these practices risk introducing out-of-distribution (OOD) values. [29] demonstrates that filling with zeros or means typically yields lower accuracy compared to using generative perturbed values, as the latter produces in-distribution values, though exceptions exist. We recommend carefully considering whether replacement values introduce new information, such as anomalies or rare pat-

Table 2: Performance results. Left: report on three types of tasks using deletion method; higher values are favored. Right: report on “OR” tasks using preservation method; lower values are favored.

Method/ Metric	AND1		AND2		AND3		OR1		OR2		Trend1		Trend2		OR1	OR2
	acc.	AUROC drop	acc.	AUROC drop	acc.	AUROC drop	acc.	AUROC drop	acc.	AUROC drop	acc.	AUROC drop	acc.	AUROC drop	AUROC drop	AUROC drop
FIT	.343	.029	.124	.369	.044	.175	.175	-.020	.177	-.020	.276	.056	.258	.015	.383	.351
WinIT	.059	.073	.061	.389	.103	.472	.022	-.020	.025	-.002	.044	.014	.092	-.004	.033	.030
FDTemp- Explainer	1.00	.110	.279	.016	.241	.136	.254	.002	.290	.002	.412	.010	.427	.000	.187	.145
FO	.373	.078	.133	.370	.103	.386	.179	-.029	.190	-.019	.073	.015	.073	.005	.384	.351
AFO	.382	.168	.124	.370	.069	.116	.173	-.033	.176	-.021	.129	.027	.089	.004	.380	.341
DeepLift	.696	.102	.079	.195	.287	.047	.217	-.041	.219	-.015	.288	.003	.272	.001	.267	.189
IG	.922	.043	.415	.501	.287	.047	.230	-.027	.231	-.017	.283	.006	.326	.025	.303	.285
GradSHAP	.902	.081	.373	.483	.287	.047	.224	-.028	.243	-.018	.274	.012	.311	.011	.306	.278
Dynamask	.118	.095	.067	.303	.103	.036	.062	-.015	.027	-.016	.051	.000	.089	.009	.094	.021
CD	.372	.074	.176	.026	.174	.052	.130	.015	.204	-.001	.158	.032	.230	.000	.092	.178

terns, as illustrated in Figure 1. To address OOD concerns, [41] proposes using reversed series as replacements, dividing the original time series into meaningful segments and reversing their temporal ordering. This approach maintains the data’s structural dynamics while providing meaningful perturbations that preserve the temporal sequence’s characteristics.

Choice of masking strategy. As discussed early, the masking strategy can introduce new information into the evaluation. As Figure 5 shows, if the prediction task is to determine the presence or absence of stairs, masking the past stair will create a new stair, thereby leaving the prediction value unchanged. This is because such TS exhibits temporal dependence that changes in earlier features can affect subsequent features. Simply masking the earlier features will not eliminate the information from those features, as it may still be retained in the subsequent features. Thereby [23] suggests three masking strategies: masking all the subsequent observations until the end, masking until changed point occurs and masking a fixed number of features.

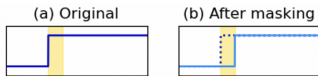


Figure 5: Masking strategy can introduce new information. After masking the current time step with the past value, the original stair disappears, but a new stair is created.

Choice of masking percentage. When evaluating feature importance, it’s crucial to report results across various masking percentages for a comprehensive comparison. While 20% or top- k features are common choices for verifying the identification of important features, relying on single percentages may be susceptible to cherry-picking. For instance, masking 90% of features will likely result in significant metric drops. A more robust approach is to plot a curve showing how the metric changes with the number of masked features, with some studies calculating the area under this curve for quantitative comparison [52].

7 Conclusion and Future Work

In this paper, we survey the methods of XAI for time series and the corresponding evaluation methods. We start from the existing XAI methods and discuss challenges for XAI for time series. We present a taxonomy of the XAI methods for

TS, and introduce them comparatively with the discussion of strengths and limitations. To conclude the paper, we summarize some potential directions worth exploring in the future.

First, the unique characteristics of time series, particularly the temporal dependency and interactions, are the key to the explanation to time series data. Although there are some progress in solving it, we believe both model-agnostic and model-specific methods can advance in this direction. A close related problem is time series segmentation for explanation. Although the uniform segmentation is simple and straightforward, automatic learning of meaningful interpretable representations is encouraged. Also, normalizing the attributions from different time lengths should be well-defined, otherwise the ranking of attribution may be altered.

Secondly, solving the OOD problem more effectively is still the focus for explaining time series. In this survey we have already introduced some methods for solving it, e.g., adopting generative models, but it is not fully solved. This challenge is complex but deserves much further investigation in the future.

Besides these theoretical problems, we observe that there is a great need of practical XAI algorithms/tools for time series tasks, especially the forecasting task, a field less explored than the time series classification tasks in the literature. Future XAI methods for forecasting should be designed to address practical requirements, such as understanding multiple feature and time step interactions (e.g., in weather forecasting, see how solar radiation affects wind, and how wind affects cloud) or finding the quantitative relationship between time segments and outcomes (e.g., how temperature fluctuations influence electricity load changes).

References

- [1] David Alvarez-Melis and Tommi S Jaakkola. On the robustness of interpretability methods. *2018 ICML Workshop on Human Interpretability in Machine Learning*. arXiv:1806.08049, 2018.
- [2] João Bento, Pedro Saleiro, André F Cruz, Mário AT Figueiredo, and Pedro Bizarro. Timeshap: Explaining recurrent models through sequence perturbations. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, pages 2565–2573, 2021.
- [3] Adrien Bibal, Rémi Cardon, David Alfter, Rodrigo Wilkens, Xiaou Wang, Thomas François, and Patrick Watrin. Is attention explanation? an introduction to the debate. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, pages 3889–3900, 2022.
- [4] Cristian Challu, Kin G Olivares, Boris N Oreshkin, Federico Garza Ramirez, Max Mergenthaler Canseco, and Artur Dubrawski. Nhits: Neural hierarchical interpolation for time series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37(6), pages 6989–6997, 2023.
- [5] Edward Choi, Mohammad Taha Bahadori, Joshua A. Kulas, Andy Schuetz, Walter F. Stewart, and Jimeng Sun. Retain: An interpretable predictive model for healthcare using reverse time attention mechanism. In *Proceedings of International Conference on Neural Information Processing Systems*, page 3512–3520. Curran Associates Inc., 2016.
- [6] Robert B Cleveland, William S Cleveland, Jean E McRae, Irma Terpenning, et al. Stl: A seasonal-trend decomposition. *J. off. Stat.*, 6(1):3–73, 1990.
- [7] Jonathan Crabbé and Mihaela van der Schaar. Explaining time series predictions with dynamic masks. In *Proceedings of the International Conference on Machine Learning*, volume 139, pages 2166–2177, 2021.
- [8] Flavio Di Martino and Franca Delmastro. Explainable ai for clinical and remote health applications: a survey on tabular and time series data. *Artificial Intelligence Review*, 56(6):5261–5315, 2023.
- [9] Finale Doshi-Velez and Been Kim. Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608*, 2017.
- [10] Mengnan Du, Ninghao Liu, Fan Yang, Shuiwang Ji, and Xia Hu. On attribution of recurrent neural network predictions via additive decomposition. In *The World Wide Web Conference*, pages 383–393, 2019.
- [11] Joseph Enguehard. Learning perturbations to explain time series predictions. In *Proceedings of International Conference on Machine Learning*. JMLR.org, 2023.
- [12] Ruth C. Fong and Andrea Vedaldi. Interpretable explanations of black boxes by meaningful perturbation. In *IEEE International Conference on Computer Vision*, pages 3449–3457, 2017.
- [13] Christopher Frye, Damien de Mijolla, Tom Begley, Laurence Cowton, Megan Stanley, and Ilya Feige. Shapley explainability on the data manifold. In *International Conference on Learning Representations*, 2021.
- [14] Ilaria Gandin, Arjuna Scagnetto, Simona Romani, and Giulia Barbati. Interpretability of time-series deep learning models: A study in cardiovascular patients admitted to intensive care unit. *Journal of biomedical informatics*, 121:103876, 2021.
- [15] Tian Guo, Tao Lin, and Nino Antulov-Fantulin. Exploring interpretable lstm neural networks over multi-variable data. In *International conference on machine learning*, pages 2494–2504. PMLR, 2019.
- [16] Long V. Ho, Melissa Aczon, David Ledbetter, and Randall Wetzel. Interpreting a recurrent neural network’s predictions of icu mortality risk. *Journal of Biomedical Informatics*, 114(C), feb 2021.
- [17] Aya Abdelsalam Ismail, Mohamed Gunady, Luiz Pessoa, Héctor Corrada Bravo, and Soheil Feizi. Input-cell attention reduces vanishing saliency of recurrent neural networks. In *Proceedings of the International Conference on Neural Information Processing Systems*, page 10814–10824. Curran Associates Inc., 2019.
- [18] Sarthak Jain and Byron C Wallace. Attention is not explanation. In *Proceedings of NAACL-HLT*, pages 3543–3556, 2019.
- [19] Neil Jethani, Adriel Saporta, and Rajesh Ranganath. Don’t be fooled: label leakage in explanation methods and the importance of their quantitative evaluation. In *International Conference on Artificial Intelligence and Statistics*, pages 8925–8953, 2023.
- [20] Tung Kieu, Bin Yang, Chenjuan Guo, Christian S Jensen, Yan Zhao, Feiteng Huang, and Kai Zheng. Robust and explainable autoencoders for unsupervised time series outlier detection. In *IEEE International Conference on Data Engineering*, pages 3038–3050. IEEE, 2022.
- [21] Colin Lea, Rene Vidal, Austin Reiter, and Gregory D Hager. Temporal convolutional networks: A unified approach to action segmentation. In *Computer Vision—ECCV 2016 Workshops: Amsterdam, The Netherlands, October 8-10 and 15-16, 2016, Proceedings, Part III 14*, pages 47–54. Springer, 2016.
- [22] Ritchie Lee, Mykel J Kochenderfer, Ole J Mengshoel, and Joshua Silberman. Interpretable categorization of heterogeneous time series data. In *Proceedings of the 2018 SIAM International Conference on Data Mining*, pages 216–224. SIAM, 2018.
- [23] Kin Kwan Leung, Clayton Rooke, Jonathan Smith, Saba Zuberi, and Maksims Volkovs. Temporal dependencies in feature importance for time series prediction. In *The International Conference on Learning Representations*, 2023.
- [24] Bryan Lim, Seran Ö Arik, Nicolas Loeff, and Tomas Pfister. Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4):1748–1764, 2021.

- [25] Yang Lin, Irena Koprinska, and Mashud Rana. Ssdnet: State space decomposition neural network for time series forecasting. In *IEEE International Conference on Data Mining*, pages 370–378. IEEE, 2021.
- [26] Zichuan Liu, Yingying Zhang, Tianchuan Wang, Zefan Wang, Dongsheng Luo, Menguan Du, Min Wu, Yi Wang, Chunlin Chen, Lunting Fan, and Qingsong Wen. Explaining time series via contrastive and locally sparse perturbations. In *The International Conference on Learning Representations*, 2024. under review.
- [27] Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, volume 30, pages 4768–4777. Curran Associates, Inc., 2017.
- [28] Han Meng, Christian Wagner, and Isaac Triguero. Feature importance identification for time series classifiers. In *IEEE International Conference on Systems, Man, and Cybernetics*, pages 3293–3298, 2022.
- [29] Han Meng, Christian Wagner, and Isaac Triguero. Explaining time series classifiers through meaningful perturbation and optimisation. *Information Sciences*, 645:119334, 2023.
- [30] W. James Murdoch, Peter J. Liu, and Bin Yu. Beyond word importance: Contextual decomposition to extract interactions from LSTMs. In *International Conference on Learning Representations*, 2018.
- [31] Yuqi Nie, Nam H Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *The Eleventh International Conference on Learning Representations*, 2023.
- [32] Boris N. Oreshkin, Dmitri Carpo, Nicolas Chapados, and Yoshua Bengio. N-beats: Neural basis expansion analysis for interpretable time series forecasting. In *International Conference on Learning Representations*, 2020.
- [33] Leonardos Pantiskas, Kees Verstoep, and Henri Bal. Interpretable multivariate time series forecasting with temporal attention convolutional neural networks. In *2020 IEEE Symposium Series on Computational Intelligence (SSCI)*, pages 1687–1694, 2020.
- [34] Leon Petrosjan and Georges Zaccour. Time-consistent shapley value allocation of pollution cost reduction. *Journal of economic dynamics and control*, 27(3):381–398, 2003.
- [35] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. “why should i trust you?”: Explaining the predictions of any classifier. In *Proceedings of the SIGKDD International Conference on Knowledge Discovery and Data Mining*, page 1135–1144, 2016.
- [36] Thomas Rojat, Raphaël Puget, David Filliat, Javier Del Ser, Rodolphe Gelin, and Natalia Díaz-Rodríguez. Explainable artificial intelligence (xai) on timeseries data: A survey, 2021.
- [37] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *IEEE International Conference on Computer Vision*, pages 618–626, 2017.
- [38] Avanti Shrikumar, Peyton Greenside, and Anshul Kundaje. Learning important features through propagating activation differences. In *Proceedings of the International Conference on Machine Learning*, volume 70, pages 3145–3153, 2017.
- [39] Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. Deep inside convolutional networks: visualising image classification models and saliency maps. In *Proceedings of the International Conference on Learning Representations*, 2014.
- [40] Chandan Singh, W. James Murdoch, and Bin Yu. Hierarchical interpretations for neural network predictions. In *International Conference on Learning Representations*, 2019.
- [41] Terty Sivill and Peter Flach. Limesegment: Meaningful, realistic time series explanations. In *International Conference on Artificial Intelligence and Statistics*, pages 3418–3433, 2022.
- [42] Mukund Sundararajan and Amir Najmi. The many shapley values for model explanation. In *Proceedings of the International Conference on Machine Learning*, volume 119, pages 9269–9278, 2020.
- [43] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *Proceedings of the International Conference on Machine Learning*, volume 70, pages 3319–3328, 2017.
- [44] Harini Suresh, Nathan Hunt, Alistair Johnson, Leo Anthony Celi, Peter Szolovits, and Marzyeh Ghassemi. Clinical intervention prediction and understanding with deep neural networks. In *Proceedings of the 2nd Machine Learning for Healthcare Conference*, volume 68, pages 322–337, 2017.
- [45] Binh Tang and David S Matteson. Probabilistic transformer for time series analysis. In *Advances in Neural Information Processing Systems*, volume 34, pages 23592–23608. Curran Associates, Inc., 2021.
- [46] Sana Tonekaboni, Shalmali Joshi, Kieran R Campbell, David Duvenaud, and Anna Goldenberg. What went wrong and when? instance-wise feature importance for time-series black-box models. In *Proceedings of the International Conference on Neural Information Processing Systems*, volume 33, pages 799–809. Curran Associates Inc., 2020.
- [47] Mattia Villani, Joshua Lockhart, and Daniele Magazzeni. Feature importance for time series data: Improving kernelshap. *International Conference on AI in Finance Workshop on Explainable Artificial Intelligence in Finance*, 2022.

- [48] Ilija Šimić, Vedran Sabol, and Eduardo Veas. Perturbation effect: A metric to counter misleading validation of feature attribution. In *Proceedings of the ACM International Conference on Information & Knowledge Management*, page 1798–1807. Association for Computing Machinery, 2022.
- [49] Qingsong Wen, Tian Zhou, Chaoli Zhang, Weiqi Chen, Ziqing Ma, Junchi Yan, and Liang Sun. Transformers in time series: A survey. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, IJCAI-23*, pages 6778–6786. International Joint Conferences on Artificial Intelligence Organization, 8 2023. Survey Track.
- [50] Sarah Wiegrefe and Yuval Pinter. Attention is not not explanation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing and the International Joint Conference on Natural Language Processing*, pages 11–20, 2019.
- [51] Junchen Yang, Ofir Lindenbaum, and Yuval Kluger. Locally sparse neural networks for tabular biomedical data. In *International Conference on Machine Learning*, pages 25123–25153, 2022.
- [52] Sen Yang, Shujian Huang, Wei Zou, Jianbing Zhang, Xinyu Dai, and Jiajun Chen. Local interpretation of transformer based on linear decomposition. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 10270–10287, 2023.
- [53] Linxiao Yang, Yunze Tong, Xinyue Gu, and Liang Sun. Explain temporal black-box models via functional decomposition. In *Forty-first International Conference on Machine Learning*, 2024.
- [54] Matthew D Zeiler and Rob Fergus. Visualizing and understanding convolutional networks. In *European Conference on Computer Vision*, pages 818–833. Springer, 2014.
- [55] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? In *Proceedings of the AAAI conference on artificial intelligence*, volume 37, pages 11121–11128, 2023.
- [56] Jianlong Zhou, Amir H Gandomi, Fang Chen, and Andreas Holzinger. Evaluating the quality of machine learning explanations: A survey on methods and metrics. *Electronics*, 10(5):593, 2021.
- [57] Tian Zhou, Ziqing Ma, Qingsong Wen, Xue Wang, Liang Sun, and Rong Jin. Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting. In *International conference on machine learning*, pages 27268–27286. PMLR, 2022.